

Seeing Thinking with the Abacus: Making Embodied Reasoning Visible Through Hybrid Sensing

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Abstract

As accuracy and performance increasingly dominate interactive systems, the temporal processes through which human understanding develops often recede into the background. Reasoning unfolds gradually through action, inspection, hesitation, and revision. In this paper, we investigate how embodied reasoning during physical calculation can be made perceptible by attending to how people act, look, and pause while interacting with a material artifact. We present a hybrid sensing prototype centered on a physical abacus that combines gaze, head movement, and bead manipulation to respond to moments of inspection as they emerge. Rather than displaying information continuously or inferring cognitive states, the system briefly surfaces numerical feedback when interaction patterns indicate checking or reflection. Findings suggest that usefulness depended less on what information was shown and more on when it appeared. We argue for interaction aligned with human sense-making rhythms, in which system responses follow observable reasoning activity without directing or accelerating it.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Interaction devices**; • **Computing methodologies** → *Artificial intelligence*; • **Applied computing** → *Education*.

Keywords

embodied cognition, hybrid sensing, gaze tracking, abacus, tangible learning, sense-making-aligned systems, reflective interfaces

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1 Introduction

At a time when interactive and AI-driven systems increasingly prioritize accuracy, speed, and performance, the temporal work through

which human understanding develops can become harder to notice. Learning rarely happens in an instant. It often unfolds through pauses, repeated inspection, small revisions, and moments of uncertainty in which intermediate states are checked and reconsidered. From perspectives in embodied cognition and human-computer interaction, reasoning is not understood as a purely internal mental operation but as something enacted through interaction with material structures that preserve intermediate configurations and make them revisitable over time. Such structures support sense-making by keeping partial states available for inspection and revision as activity unfolds, rather than compressing reasoning into immediate outcomes. In this work, we ask how an interactive system might attend to these temporal dynamics of embodied reasoning without accelerating interaction, providing instruction, or inferring internal cognitive states. As a point of reference, we draw on the abacus as a calculating artifact whose use consists of visible, reversible physical actions. Through bead manipulation, intermediate numerical states remain materially present and open to inspection, allowing calculation to proceed as an ongoing and revisitable process.

We present a hybrid sensing prototype in which learners manipulate a physical abacus while wearing a head-mounted device. At certain moments, the numerical value implied by the current bead configuration becomes briefly visible as a spatially anchored holographic overlay above the physical abacus and then fades out again. Crucially, this overlay is not continuously available and cannot be invoked through explicit input. Instead, a lightweight orchestration logic relates gaze behavior, head movement, and bead manipulation over time to determine when the value should remain absent and when it may briefly appear. The contribution of this work is therefore not the visualization of a numerical value per se, but an interaction strategy that withholds numerical information by default and surfaces it only in situational moments of inspection. Through an exploratory study with university students, we examine how the timing and situational relevance of these momentary appearances are experienced during physical calculation. We argue that such sense-making-aligned interaction can support reflection by making reasoning perceptible, while preserving the open-ended unfolding of sense-making. **Research Question:** How can a hybrid sensing system respond to temporal patterns of embodied activity during calculation without inferring understanding or providing assistance?



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2 Background and Related Work

2.1 Embodied Reasoning and Epistemic Things

A long tradition in cognitive science and HCI emphasizes that human reasoning is distributed across mind, body, and environment. Rather than operating solely on internal representations, people routinely rely on perceptual, bodily, and material resources to support understanding. From this perspective, cognition unfolds through situated action, with external structures actively shaping what can be perceived, manipulated, and reflected upon. Several theoretical accounts highlight the epistemic role of material artifacts in this process. Clark characterizes cognition as “leaky,” noting how external representations are recruited into cognitive processes rather than merely consulted [4]. Dourish and Tversky similarly argue that meaning and reasoning emerge through embodied, situated interaction, in which perception and action are inseparable from interpretation [5, 17]. Rheinberger’s notion of epistemic things foregrounds objects whose properties become knowable through manipulation, while Papert’s objects-to-think-with illustrate how acting on artifacts supports reflection by making intermediate states persistent and inspectable [13, 14].

The abacus, as a physical calculating tool, exemplifies such an epistemic artifact. Through material manipulation, intermediate calculation states remain visible and revisitable, allowing learners to externalize partial results, detect inconsistencies, and stabilize emerging strategies. Empirical work on abacus use shows that sustained engagement reshapes calculation practices and supports the development of embodied representations, including the internalization of spatial and procedural structures [16, 18]. More broadly, research on physical manipulatives and tangible interaction demonstrates that material form and spatial configuration influence how numerical and conceptual understanding develops [10, 12]. Foundational work on tangible interfaces further emphasizes hybrid approaches that preserve material engagement rather than replacing it with abstract digital representations [7, 8]. Together, this body of work underscores that understanding does not emerge despite material engagement, but through it. Artifacts that preserve intermediate states support reasoning by making processes legible over time instead of compressing them into outcomes.

2.2 Gaze, Multisensory Sensing, and Learning Technologies

In parallel, HCI and learning technologies have extensively explored eye gaze and multimodal sensing as sources of information about user attention, intention, or cognitive effort. Gaze has been employed both as an explicit interaction modality, for example through gaze gestures [6], and as an implicit signal for estimating visual attention from head pose [1], recognizing activities [2], or predicting learner states such as confusion or engagement [9]. In immersive and spatial computing contexts, gaze analysis has been extended to three-dimensional environments, enabling fine-grained mapping of visual attention in relation to virtual and physical structures [3]. Across these approaches, multimodal data are commonly treated as inputs for inference or control. Fixations, regressions, and dwell times are abstracted into features used to classify internal states, predict outcomes, or adapt system behavior. Survey work on

gaze and multimodal inference in HCI and learning technologies highlights this emphasis on prediction, efficiency, and optimization [15, 19].

Recent research, however, emphasizes temporality as a critical dimension for interpreting gaze and interaction data. Patterns such as post-action pauses, gaze regressions toward prior configurations, and sustained fixations have been associated with moments of verification, hesitation, or consolidation [2, 3]. In embodied learning contexts, these temporal patterns often arise immediately after material transformations, when learners shift from acting to inspecting intermediate states [10]. While prior work has leveraged such patterns to infer latent cognitive states [11], we deliberately refrain from treating them as evidence of understanding or confusion. Rather than operationalizing these behaviors as indicators of internal states, we treat gaze, bodily movement, and material manipulation as observable temporal cues. From this perspective, multimodal sensing is not used to infer what a learner knows, but to attend to how interaction unfolds over time. In our approach, multisensory input informs restrained system timing by helping decide when a brief response may be situationally appropriate, while avoiding instructional adaptation, prediction, or claims about understanding.

3 Prototype and Technical Setup

We developed a hybrid sensing prototype that augments interaction with a physical abacus by making aspects of calculation situationally visible. The system is intentionally non-directive: it does not provide instructions, alter the material configuration, or infer internal cognitive states. Instead, it relates multiple observable signals to how interaction unfolds over time during physical calculation. As illustrated in Figure 1, the prototype integrates three parallel sensing streams: (1) bead movement on the physical abacus, (2) gaze behavior, and (3) head position and orientation. Bead positions are captured using an overhead camera and a task-specific computer vision pipeline sufficient to track bead configurations during interaction. Detected bead movements are continuously mirrored in an internal digital representation of the abacus. While this representation remains invisible to the learner, it provides the spatial frame of reference required to relate gaze behavior and head orientation to specific bead configurations over time. The numerical value implied by the current bead configuration can be rendered as a spatially anchored holographic overlay positioned above the physical abacus. This numerical overlay is not shown by default and cannot be invoked explicitly; all changes to the calculation state result exclusively from bead manipulation by the learner. The timing of the overlay is determined by the temporal relation between gaze behavior, bead manipulation, and head orientation. Rather than using single-signal thresholds, the system relies on conservative combinations of temporal cues that must persist or recur before a response occurs. System responses are driven primarily by gaze behavior unfolding over time. Single fixations, brief glances, or continuous visual attention alone do not lead to the appearance of the numerical overlay. Instead, the system responds only to temporally extended gaze patterns, such as repeated regressive fixations, sustained fixations on the same bead, or visual attention that returns to a recently manipulated region following a pause.

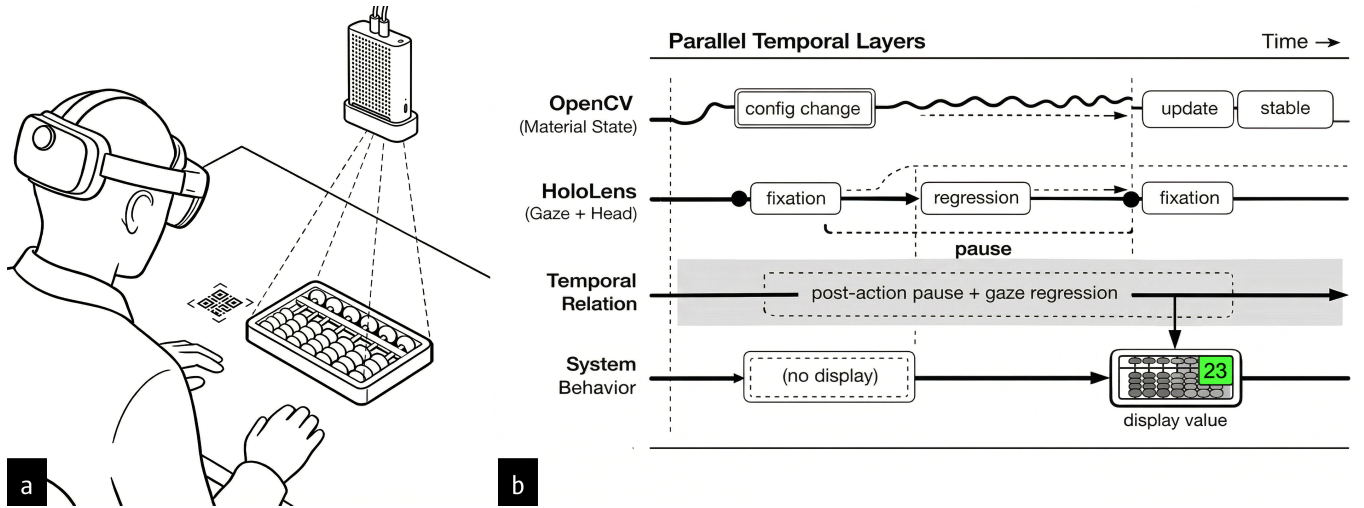


Figure 1: (a) Participant wearing a HoloLens; a visual marker beside the physical abacus provides a spatial reference for registering the digital overlay, while an overhead camera tracks bead movements. (b) Parallel temporal layers showing material state (bead movement), gaze and head behavior, and their temporal relation. A lightweight orchestration layer relates these streams over time to determine when the numerical value implied by the bead configuration is briefly displayed.

Bead manipulation provides a temporal frame for interpreting gaze: active movement suppresses any overlay, while pauses following configuration changes allow gaze behavior to be evaluated in relation to the newly established bead state. Head pose is used as a coarse contextual filter to ensure that gaze events are considered only when participants are oriented toward the abacus. Across all conditions, responses are conservatively rate-limited so that repeated checking behavior does not result in frequent or predictable appearances of the overlay.

4 Exploratory Study

This exploratory, formative study examined how participants perceived the behavior of the hybrid sensing prototype while calculating with a physical abacus. The study did not aim to assess learning outcomes, correctness, or performance. Instead, it focused on how the timing and situational relevance of the system’s brief responses were experienced during interaction. Twelve university students (aged 21–28) without prior experience using an abacus participated in the study. The study was designed to probe subjective impressions of the system’s behavior and to explore whether the intermittent numerical overlay was perceived as understandable, appropriately timed, and supportive of reflection. Given this exploratory focus, the study did not aim for statistical generalization.

4.1 Procedure and Data Collection

Participants interacted with a physical abacus while wearing a Microsoft HoloLens. After eye-tracking calibration, they were introduced to the abacus as a calculating artifact, without explanation of its operational principles or of the system’s behavior. Before starting the tasks, participants were given a short period to freely explore the setup and familiarize themselves with the material configuration. The study consisted of ten arithmetic tasks involving addition and subtraction. Early tasks were designed to be solvable

mentally, while later tasks favored the use of the abacus. Participants were given unlimited time for each task and were explicitly told that speed was not important. They were asked only to set their final result on the abacus. During interaction, the system rendered the current numerical value as a spatially anchored holographic overlay above the physical abacus. Participants were not informed about when or why this value would appear. The overlay did not provide hints, corrections, or confirmations. Each session lasted approximately 25 minutes. After completing the tasks, participants responded to eleven questionnaire items. For reporting, we focus on four items directly related to the research question, addressing perceived timing, interpretability, anticipation, and support during moments of uncertainty (Fig. 2). This was followed by a short semi-structured interview focusing on how the timing of the overlay was perceived during interaction.

5 Preliminary Observations

The following observations summarize participants’ impressions from the questionnaire responses and post-task interviews. They do not assess learning outcomes or prototype effectiveness, but describe how participants experienced the system’s behavior while calculating with the physical abacus. Participants quickly understood the role of the holographic numerical overlay. During the initial exploration phase, they recognized that the displayed number reflected the numerical value implied by the bead configuration they had physically set on the abacus. The appearance of the overlay was not perceived as random or disruptive, but as understandable and situationally related to their ongoing activity. Participants reported primarily using the displayed value to check their current calculation state and to notice moments of uncertainty, rather than as guidance or correction. Several participants noted that the overlay tended to appear primarily in moments when they felt uncertain about the current bead configuration or the value it represented.

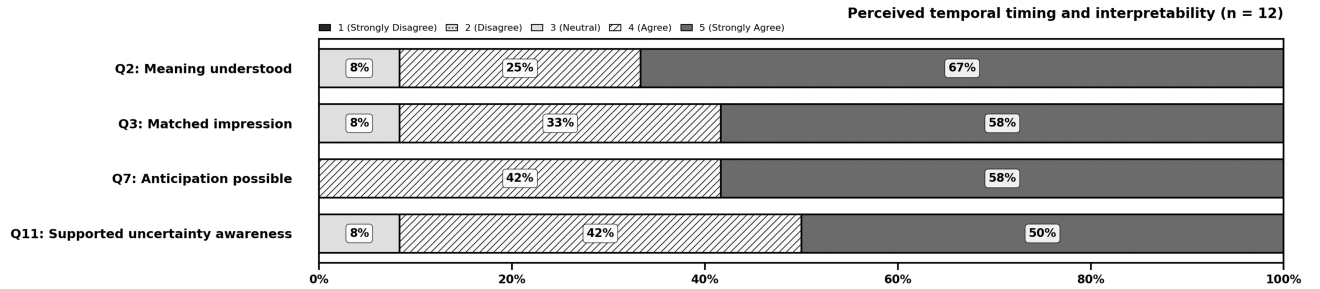


Figure 2: Questionnaire ratings on perceived timing and interpretability (n = 12). Four of eleven questionnaire items directly addressing the research question are shown (Q2, Q3, Q7, Q11). At least 92% of responses were neutral or positive (50–67% “strongly agree”); remaining items showed comparable distributions (median ≥ 3).

Conversely, when they felt confident about their calculation, the absence of the overlay was perceived as appropriate and unsurprising rather than as missing feedback. These accounts suggest that participants interpreted not only the presence of the overlay, but also its absence, as meaningful in relation to their own sense-making process. With repeated use, some participants described a growing sense of confidence when working with the abacus, which they attributed to the system’s calm and non-directive behavior. Across most sessions, participants showed little head movement during focused calculation. Visual attention remained oriented toward the bead configuration and hand movements. Only one participant briefly turned their head away from the abacus during a moment of deliberation before returning to the task. These observations indicate that, in this setting, interaction remained closely coupled to the material structure, with relevant cues expressed mainly through pauses, gaze behavior, and bead manipulation rather than through larger bodily reorientation.

5.1 Observed Limitation: Reasoning Beyond Visual Engagement

We also observed episodes in which some participants appeared to rely on mental calculation strategies alongside material interaction—for example by briefly closing their eyes or looking away while continuing to reason. Because the current system responds to gaze behavior and head orientation in relation to the abacus, such phases fall outside its present scope. This highlights a limitation of the current approach and points toward future work on how sense-making-aligned systems might account for reasoning episodes that temporarily occur beyond direct visual engagement with a material artifact.

6 Discussion and Design Implications

Our exploratory study suggests that embodied reasoning during physical calculation becomes perceptible when perception, action, and material manipulation are considered in their temporal relation. Rather than exposing or inferring internal cognitive states, the prototype made moments of checking and reflection visible by responding to how interaction unfolded over time. Participants did not experience the system’s behavior as instructional or corrective. Instead, they valued that numerical feedback appeared in

moments when they were already inspecting or reconsidering a bead configuration they had produced themselves. Importantly, participants also interpreted the absence of feedback as meaningful: when reasoning proceeded fluently or confidently, the lack of an overlay was perceived as appropriate restraint rather than as missing support. The usefulness of the system therefore depended less on the information displayed and more on when it appeared. These findings point toward interactive systems that respond to observable temporal patterns of embodied activity—such as pauses following material manipulation, repeated inspection of the same configuration through gaze regressions, or brief phases in which bead movement slows or stops—while preserving the open-ended unfolding of reasoning. Designing for such temporal sensitivity requires systems to remain deliberately restrained, resisting the impulse to continuously display information or to intervene whenever uncertainty might be inferred.

6.1 Toward Sense-Making-Aligned Interaction

We refer to this approach as sense-making-aligned interaction. The term describes systems whose responses are aligned with observable sense-making activity as it unfolds through interaction, rather than with inferred cognitive states or instructional objectives. Sense-making-aligned interaction is not concerned with providing answers or guidance, but with attending to moments in which people pause, inspect, and re-check intermediate states during ongoing activity. In our prototype, responses did not result from isolated fixations, regressions, or movements. Instead, they emerged from temporally extended patterns that accumulated over time. Early or transient signals were deliberately ignored, and repeated checking behavior did not automatically lead to repeated responses. This conservative orchestration helped prevent the overlay from becoming a shortcut or a reliable trigger, while still supporting moments of self-checking and orientation. From this perspective, multisensory sensing is used to align system response timing with human sense-making rhythms, not to optimize performance or predict internal states. Participants’ accounts suggest that trust and relevance arose primarily from temporal fit: the overlay was perceived as helpful when it coincided with participants’ own moments of checking, while its absence during fluent understanding was experienced as appropriate.

7 Conclusion and Future Work

In this Late-Breaking Work, we investigated how embodied reasoning during physical calculation can be made perceptible by attending to how people act, look, and pause while interacting with a material artifact. Using a hybrid sensing setup centered on a physical abacus, we explored how multisensory data can be used to respond to moments of sense-making as they become observable in interaction, without inferring internal cognitive states or steering action. Our findings show that the perceived usefulness of the system depended less on what information was displayed and more on when it appeared. Participants described the numerical overlay as helpful when it coincided with their own moments of checking or reflection, and as unnecessary when reasoning proceeded smoothly. In this way, the system supported reflection by aligning with learners' sense-making rhythms rather than directing or accelerating calculation. Future work will extend this approach to collaborative scenarios in which multiple learners interact with shared physical artifacts, examining how multisensory sensing can be coordinated to make collective moments of checking or reflection perceptible. Additional sensing modalities, such as coarse posture cues, may further contextualize gaze and material interaction. To support reproducibility and future quantitative analyses, we plan to release the Unity project, including code, thresholds, rate limits, and example logs, alongside the published paper.

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